

Anemia

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Fall 2025



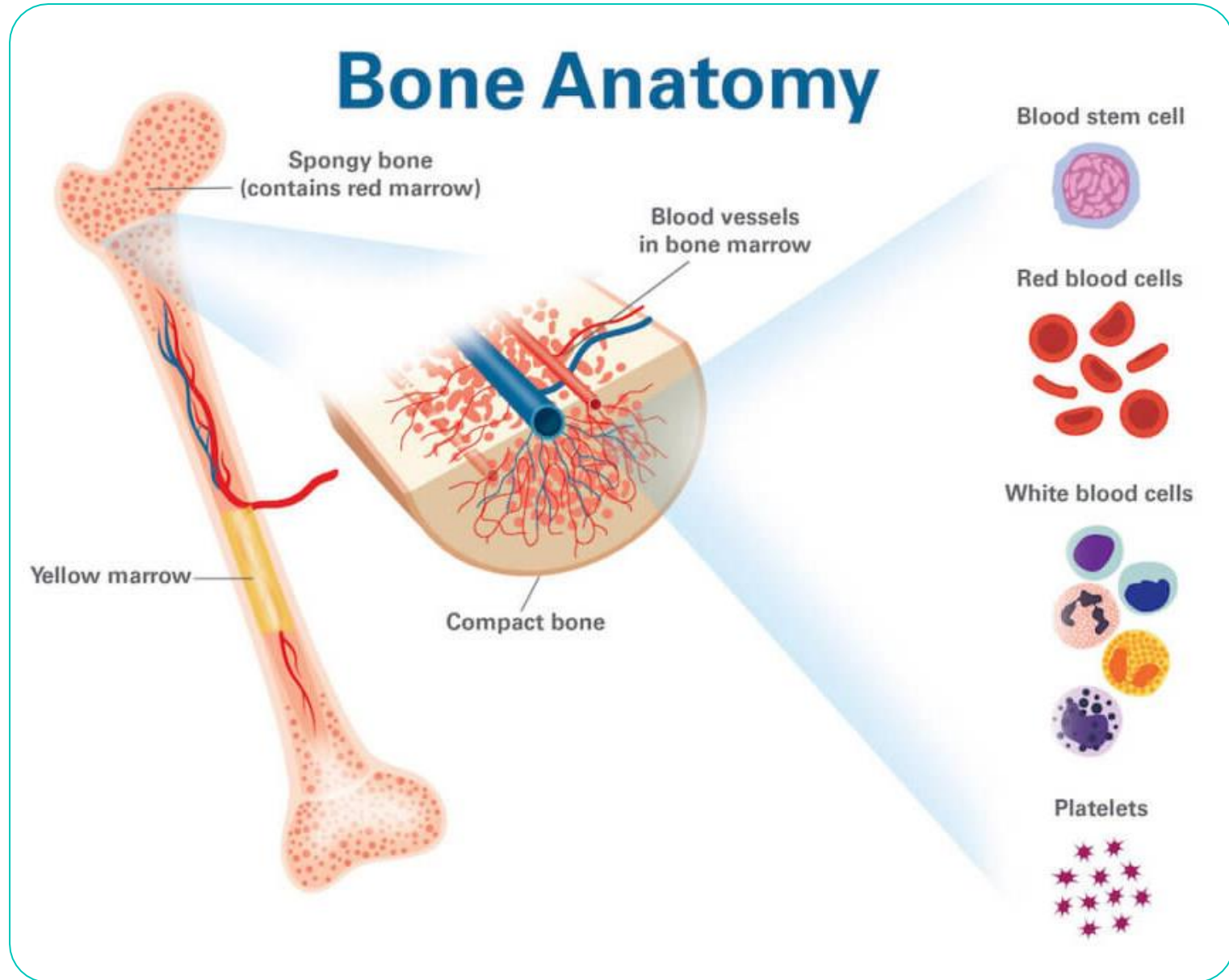
Let's Review

Blood cells are created in the Bone Marrow

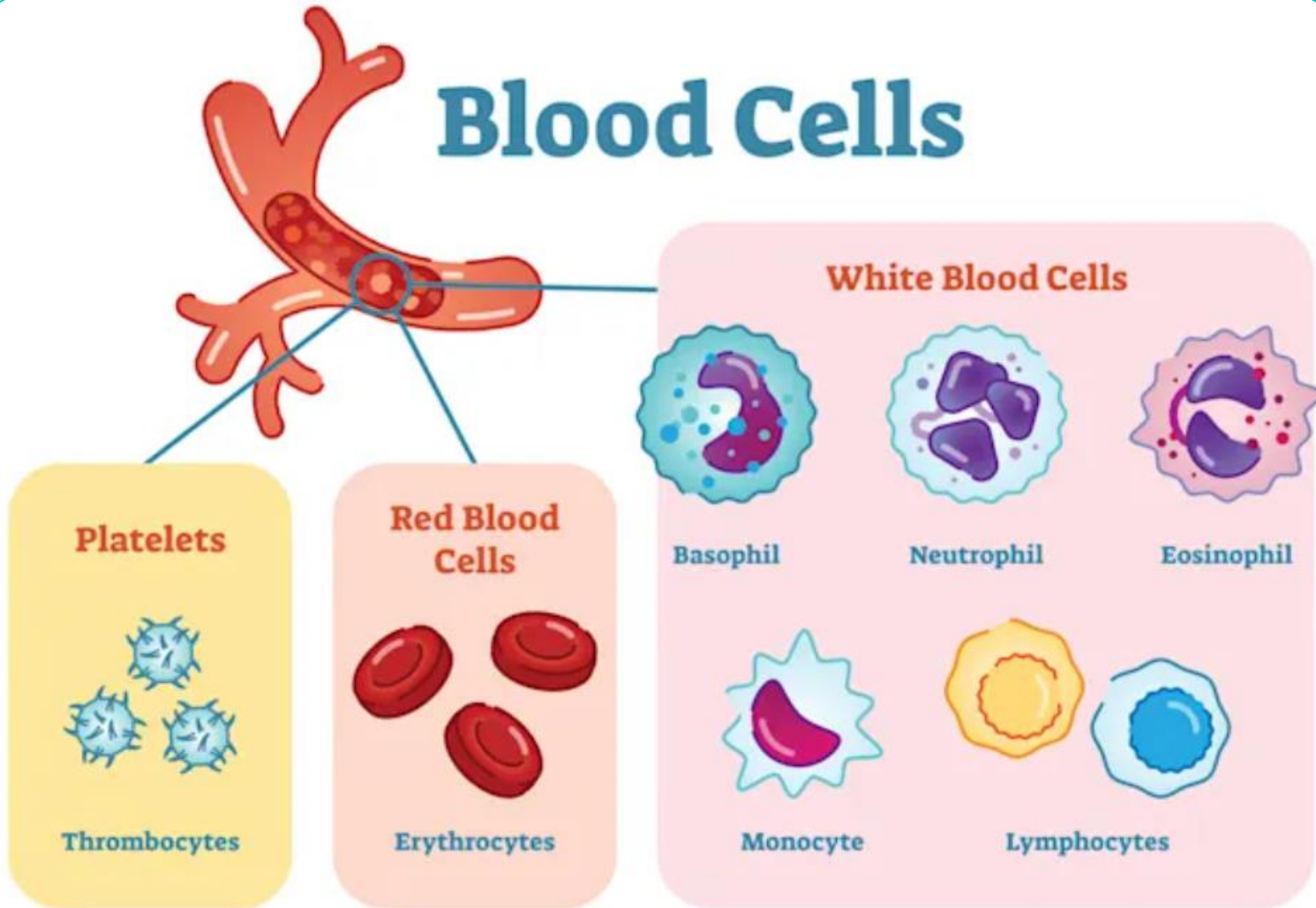
When the Kidneys sense blood levels are low, they send erythropoietin to the Bone Marrow signaling it to produce more blood cells.

Function of Blood:

- Transports O₂ & nutrients to cells
- Removes CO₂ & other waste products
- Coagulation



Blood Cells



Remember:

- **Plasma** ($\approx 55\%$ of blood volume; made of H_2O , electrolytes, proteins, nutrients, hormones, waste products)
- **White Blood Cells (WBCs)** - protection/immunity.
- **Platelets (Plts)** – blood clotting.
- **Red Blood Cells (RBCs)** - O_2 & CO_2 delivery/removal...

Continued...

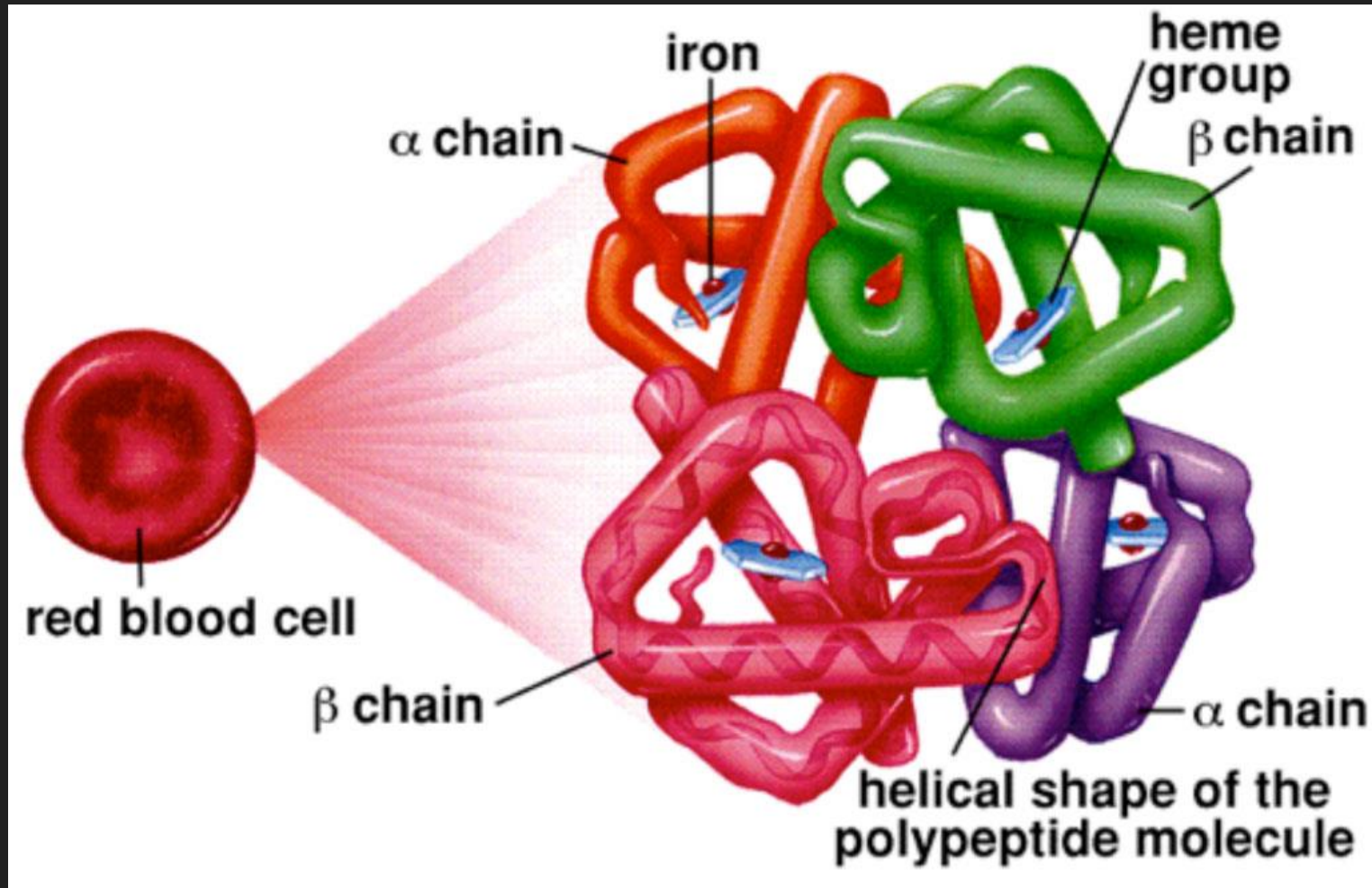
Blood A&P continued...



- **Red Blood Cells** – aka RBCs, Erythrocytes

- Made of mostly **Hemoglobin** – a protein that carries oxygen (via iron connections); also gives RBCs/blood the red color

- Large surface area for efficient gas exchange & Biconcave disc shape helps facilitate movement through blood vessels



Anemia Explained



- **Definition:** Anemia is a condition where there is a deficiency in the quantity or quality of RBCs (thus Hemoglobin & those Iron connectors used for gas transport & exchange.) Leading to reduced oxygen delivery to tissues.
- **Types:** acute/chronic blood loss, hypoproliferative , and hemolytic.
- **Impact:** Decreased oxygen delivery causes fatigue, pallor, SOB, and other systemic symptoms.

Types of Anemia

- **Blood Loss:** Acute or chronic bleeding reduces RBC count, leading to anemia.
- **Hypo-proliferative:** Decreased RBC Production is often due to bone marrow dysfunction or nutrient deficiencies (iron, vitamin B12, folic acid). This is seen in anemia of chronic disease.
- **Hemolytic Anemia:** RBC Destruction (Hemolysis) occurs when RBCs are prematurely destroyed in the bloodstream or spleen.

Global Signs & Symptoms of Anemia

Fatigue, weakness, malaise

Pallor or jaundice

Cardiac, GI, neuro, respiratory symptoms

Tongue changes

Nail changes

Angular cheilitis

Pica

Diagnostic Tests

Hemoglobin – protein that contains iron that carries Oxygen (Part of H&H or CBC)

Hematocrit – percent of blood volume made of RBCs (Part of H&H or CBC)

Reticulocyte count – Percent of immature RBCs in circulation.

Iron studies – Serum iron, Ferritin, TIBC. Specific studies.

Vitamin B₁₂ (cobalamin) – necessary in DNA synthesis & myelin production

Folate (aka B₉) – necessary in RBC DNA synthesis

Haptoglobin – indicates destroyed RBCs

Erythropoietin – hormone that stimulates bone marrow to make RBCs

Bone marrow aspiration – evaluates function including RBC production

RBC Indices (next slide)

Diagnostic Tests

- **RBC Indices:** **MCV helps determine type of anemia**
- **MCV** (Mean Corpuscular Volume): **average size of RBC**
 - **Microcytic** (small/low MCV): iron deficiency
 - **Macrocytic** (large/high MCV): B12 or Folate deficiency
 - **Normocytic** (normal MCV): acute blood loss, chronic disease
- **MCH** (Mean Corpuscular Hemoglobin): **amount of Hgb per RBC**
- **MCHC** (Mean Corpuscular Hemoglobin Concentration): **average concentration of Hgb in RBCs**

Global Treatment

- Correct or control the cause!!
- Transfusion of packed RBCs
- Treatment specific to the type of anemia
 - Dietary therapy
 - Iron or vitamin supplementation: iron, folate, B₁₂
 - Transfusions
 - Immunosuppressive therapy (if part of underlying cause)

Acute/Chronic Blood Loss Anemia

Acute or Chronic bleeding reduces RBC count, leading to anemia.

- Can be from injury
- External/Internal bleeding
- GI bleeding (upper/lower)
- Menstrual bleeding
- Surgical losses



Hypo-proliferative Anemia

- **Hypo-proliferative:** Decreased RBC Production is often due to bone marrow dysfunction or nutrient deficiencies (iron, vitamin B12, folic acid). This is seen in anemia of chronic disease

**HYPO – less/under & PROLIFERATE – create
“under-creation” of RBCs by bone marrow**

Hypo-proliferative Anemia

- **Iron deficiency anemia**
- Anemia in renal disease
- Anemia of inflammation
- Aplastic anemia
- **Megaloblastic anemia**
 - **Folic acid deficiency**
 - **Vitamin B₁₂ deficiency**



Iron Deficiency Anemia

**Necessary for
HGB Synthesis!**

PATHO:

- Decreased iron results in decreased RBC production
- Intrinsic Factor found in GI tract allows absorption of B12

ETIOLOGY:

- Low iron due to inadequate intake, increased demand (pregnancy), or chronic blood loss (GI bleed, menstruation)

S&S

- General anemia S&S, Glossitis (smooth shiny tongue), Pica, brittle nails, angular cheilitis

Iron Deficiency Anemia

**Iron + Vit C to
increase absorption!!**

TREATMENT:

- Iron supplementation
 - Via diet
 - Tablet form
 - IV Infusion
 - Liquid Suspension (better for kids, but will stain teeth so always use a straw)
- Blood product transfusion (PRBCs, plasma)

ANIMAL PRODUCTS		PLANT PRODUCTS	
			
Oysters 4mg/100g	Lean Beef 2.2mg/100g	Red Kidney Beans 1.2mg/50g	Lentils 6.7mg/100g
			
Whole Eggs 1.9mg/100g	Mussels 3mg/100g	White Quinoa 3.6mg/100g	Rolled Oats 3.5mg/100g
			
Chicken Liver 9.8mg/100g	Kangaroo 3mg/100g	Pumpkin Seeds 2mg/20g	Spinach 1.6mg/50g

Pernicious Anemia

PATHO:

- B12 deficiency due to lack of **intrinsic factor**
- Intrinsic Factor found in GI tract allows absorption of B12

ETIOLOGY:

- GI pathophysiology of destruction of parietal cells affects production of intrinsic factor
- Those with GI issues at increased risk including bariatric surgery, Chron's Disease, & Celiac's.

S&S

- General anemia S&S, Glossitis (smooth, beefy red tongue),
- *Mood changes (irritability &/or depression), *paresthesia, *ataxia

*Pernicious & Folate def. anemias have similar symptoms these neuro are specific to B12

Pernicious Anemia

TREATMENT:

- Begins with oral B12 supplementation (diet & tablet form)
- Change to IM B12 supplementation as there is less and less intrinsic factor to absorb B12.
- Monitor B12 levels

HIGH VITAMIN B12 FOODS



liver
63 mg/77 g serve



oyster
29 mg/165 g serve



mussels
19 mg/73 g serve



tuna
10 mg/96 g serve



heart
8 mg/71 g serve



ribeye filet
8 mg/220 g serve



mackerel
7 mg/92 g serve



lamb chops
6 mg/180 g serve



ground beef
5 mg/160 g serve



salmon
5 mg/120 g serve



lamb roast
5 mg/125 g serve



turkey breast
5 mg/260 g serve

Folic Acid Deficiency Anemia



kidney beans
290 mcg/210 g serve



lentils
270 mcg/125 g serve



liver
220 mcg/77 g serve



black beans
170 mcg/92 g serve



spinach
170 mcg/100 g serve



asparagus
150 mcg/85 g serve



lettuce
130 mcg/71 g serve



chickpeas
130 mcg/62 g serve



broccoli
130 mcg/100 g serve



cod liver oil
120 mcg/37 g serve



edamame
110 mcg/38 g serve



Brussels sprouts
90 mcg/110 g serve

PATHO:

- Folic acid deficiency due to decreased intake or increased need

ETIOLOGY:

- Decreased intake (esp in alcoholism) or increased need (pregnancy, growth spurts)

SIGNS & SYMPTOMS

- Same as pernicious anemia except the mood/neuro symptoms

TREATMENT

- Folic acid supplementation

Hemolytic Anemia

Anemia caused by premature destruction of RBCs.

Types:

- **Sickle cell disease**
- Thalassemia
- Glucose-6-phosphate dehydrogenase deficiency
- Immune hemolytic anemia (including transfusion reaction)
- Hereditary hemochromatosis

Hemolytic Anemia

PATHO:

- Anemia caused by premature destruction of RBCs.
- Because the bone marrow cannot *keep up*, there's less RBCs

ETIOLOGY:

- Etiology will depend on type

SIGNS & SYMPTOMS: (vary by type, but generally)

- Increased *dead* RBCs → increased bilirubin → Jaundice & dark urine & Splenomegaly
- Because there's less RBCs, same global S&S, increased reticulocyte count

Hemolytic Anemia



TREATMENT:

- Fix underlying cause
- Blood product transfusion (PRBCs, plasma)
- Anemia caused by premature destruction of RBCs.

IMPORTANT TO NOTE:

- Most blood transfusion reactions are hemolytic (you will cover that more later in the program)

Sickle Cell Anemia

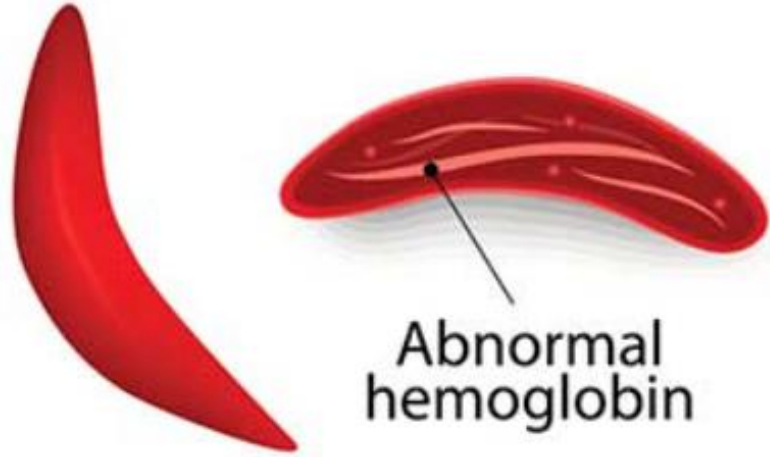
PATHO:

- Genetic disorder where RBCs sickle under stress due to abnormal Hgb
- The misshapen cells leads to hemolysis and blood vessel occlusion; causes pain & potentially organ damage; fragile cells have a much shorter lifespan than regular RBCs

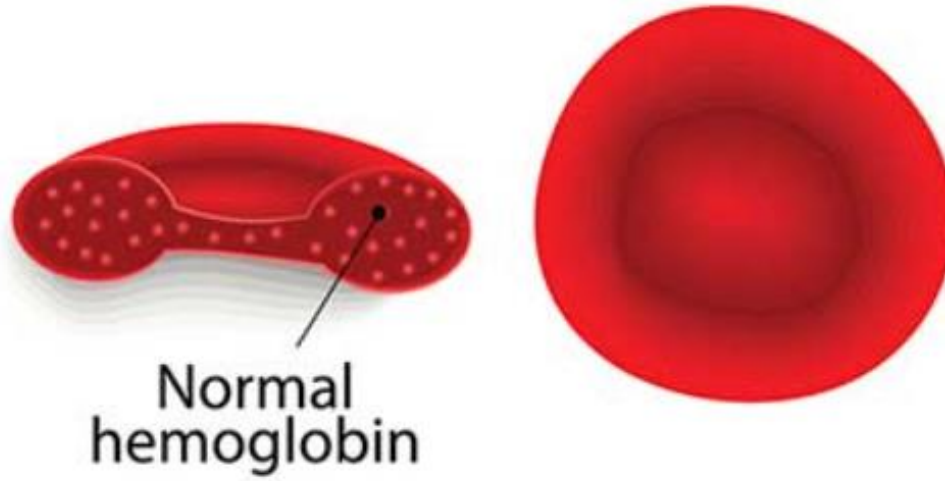
ETIOLOGY:

- Mother and Father must **both** carry the gene & child will have 25% chance of disease.

Sickle cell



Normal red blood cell



Sickle Cell Anemia: *The Chronic Condition*

SIGNS & SYMPTOMS:

- Fatigue, pallor/jaundice, SOB, impaired wound healing, growth delays, splenomegaly, frequent infections, joint pains

TREATMENT / PREVENTION:

- Stay hydrated
- Immunizations to promote wellness
- Supplements to help RBC production (iron, B12, folate)
- Hydroxyurea (helps reduce sickling episodes)
- Focus on preventing Sickle Cell Crisis by avoiding triggers (discussed on next slide)

****Must also focus on psychological aspect of the disease****

Sickle Cell Anemia: *Sickle Cell Crisis*



TRIGGERS:

- Dehydration - Low O₂ - Extreme Hot or Cold temperatures - Infection - Extreme Stress/Exertion - High Altitude

SIGNS & SYMPTOMS:

- Severe pain, fatigue, pallor/jaundice, SOB/dyspnea, occlusive events, worsening anemia

TREATMENT:

- Aggressive hydration (decrease blood viscosity)
- Pain Management (Opiates)
- Rest & recovery
- Supplemental O₂
- Blood transfusion if severe anemia
- Antibiotics if infection



Thank you, Chrissy